Reinforcement Learning

Temporal-Difference Learning (RLbook 6)

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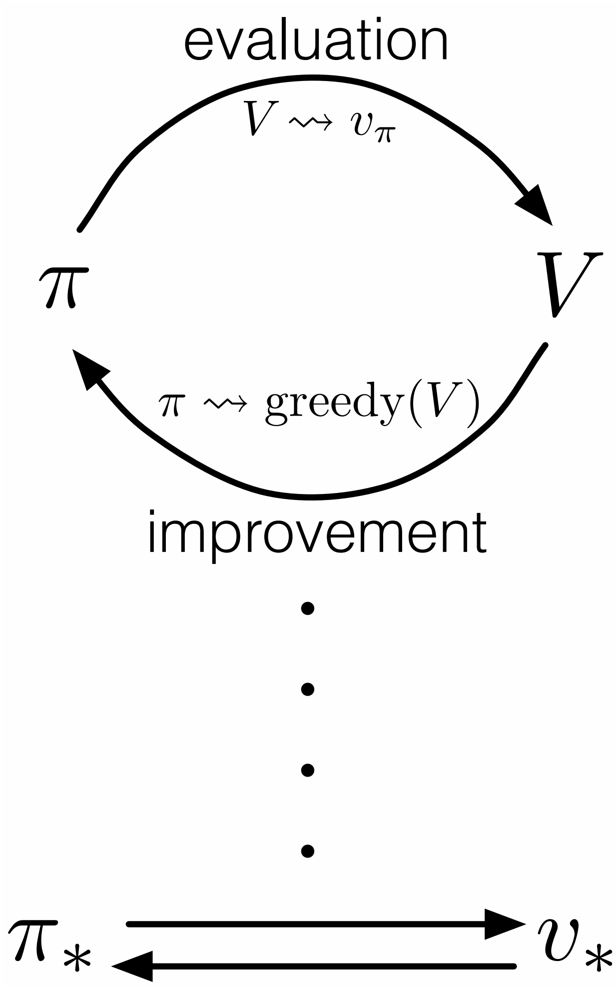
1 Temporal-Difference Learning

- DP Review
  - Dynamic programming
  - Generalized policy iteration (GPI)
- Model-free control
  - Monte Carlo control
  - Temporal-difference learning

2 Dynamic Programming

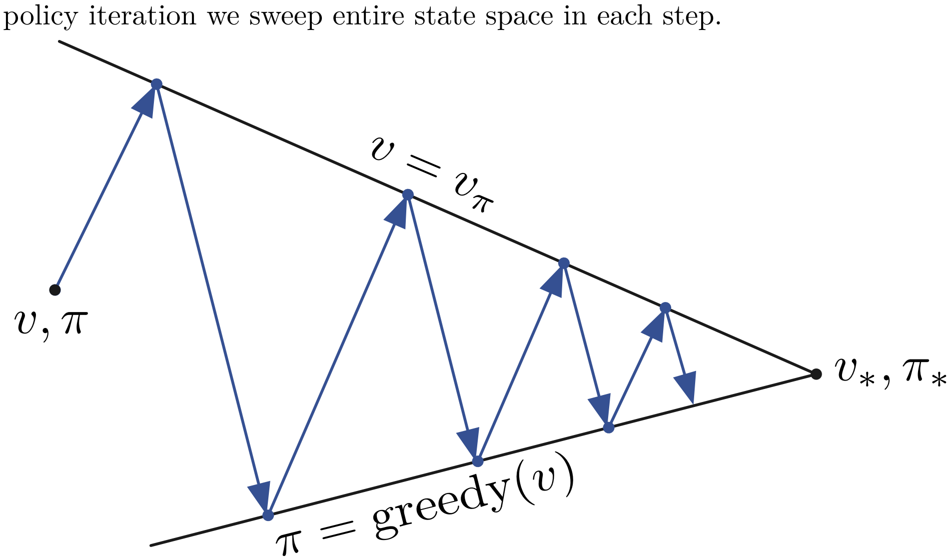
Dynamic programming algorithms, as well as RL algorithms in general, contain two phases (combined in value iteration):

- Prediction: estimate the value function
- Control: computing or approximating optimal policies



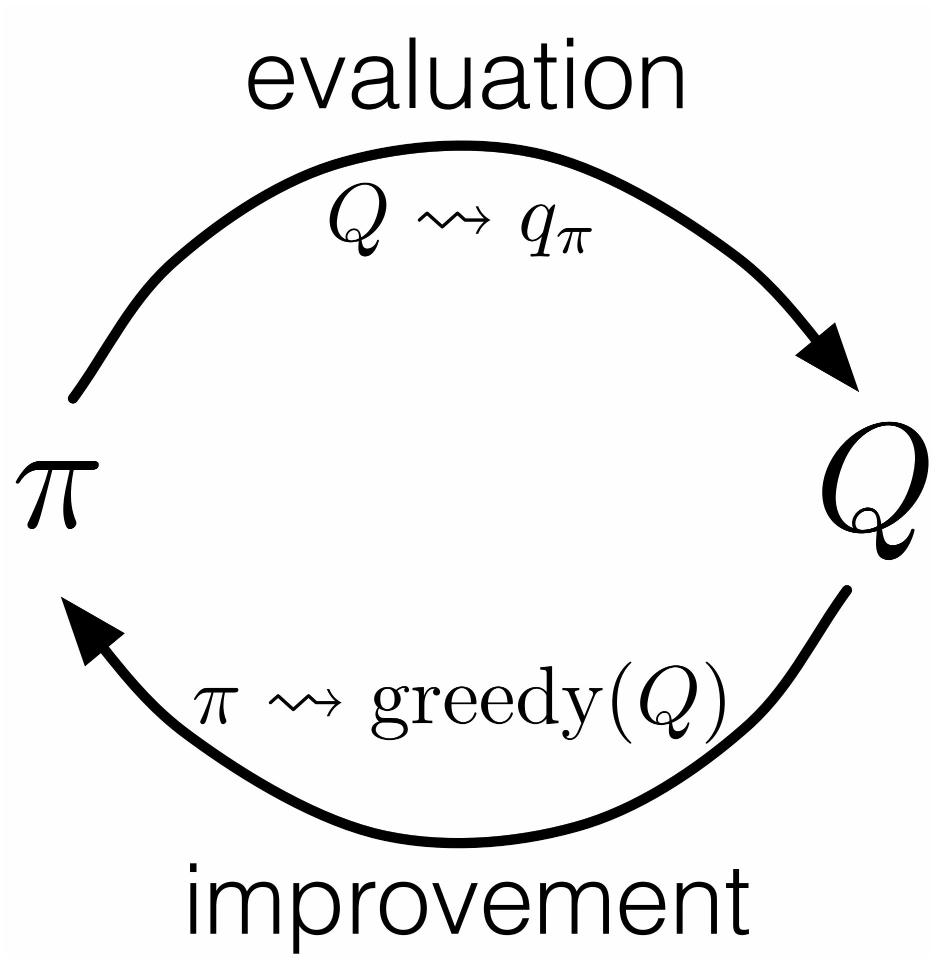
3 Generalized Policy Iteration

In policy iteration we sweep entire state space in each step.



In generalized policy iteration (GPI) we interleave prediction and control at arbitrary granularity.

- As long as we visit every state, still assured of convergence.



4 Monte Carlo Methods

Monte Carlo methods sample and average returns for each state–action pair.

- Generate a set of episode samples by following a policy π
- For each visit to a state s, find the first visit to s, then average all the returns following the first visit to s to estimate the value of s

Combination of dynamic programming and Monte Carlo ideas.

- Like Monte Carlo, learn directly from experience without a model of the environment.
- Like dynamic programming, update estimates based in part on other learned estimates, without waiting for a final outcome (bootstrap).

5 TD Prediction

Use experience following a policy to update estimate of V, namely v_{pi}.

Monte Carlo methods wait until the return following the visit is known, then use that return as a target for $V(S_t)$:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[G_t - V(S_t) \right] \tag{6.1}$$

TD methods only until the next time step. At time $t+1$ they immediately form a target and make a useful update using the observed reward R_{t+1} and the estimate $V(S_{t+1})$

A simple TD method makes the update:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right]$$

immediate on transition to S_{t+1} and receiving R_{t+1}.

- Target for Monte Carlo update is G_t.
- Target for TD update is $R_{t+1} + \gamma V(S_{t+1})$

6 Tabular TD(0) Algorithm

One-step TD is called $TD(0)$, which is a special case of $TD(\lambda)$ and n-step methods.

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\mathit{terminal}) = 0$

Loop for each episode:
  Initialize S
  Loop for each step of episode:
    $A \leftarrow$ action given by π for S
    Take action A, observe R, S'
    $V(S) \leftarrow V(S) + \alpha\big[R + \gamma V(S') - V(S)\big]$
    $S \leftarrow S'$
  until S is terminal

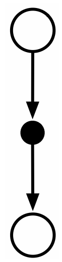
7 TD Error

Recall TD update:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right]$$

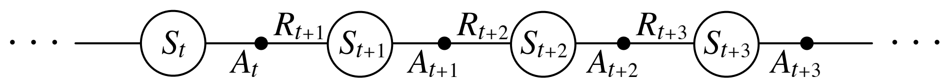
The quantity in brackets is called **TD error** – the difference in the estimate value of S at time t and $t+1$:

$$\delta \doteq R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$



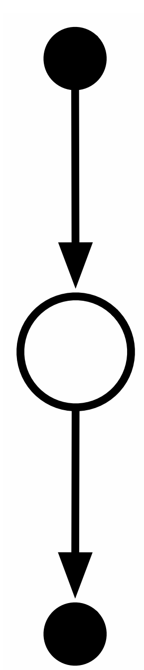
TD(0)

8 Sarsa: On-policy TD Control



Sarsa update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(S_t, A_t)\right]$$



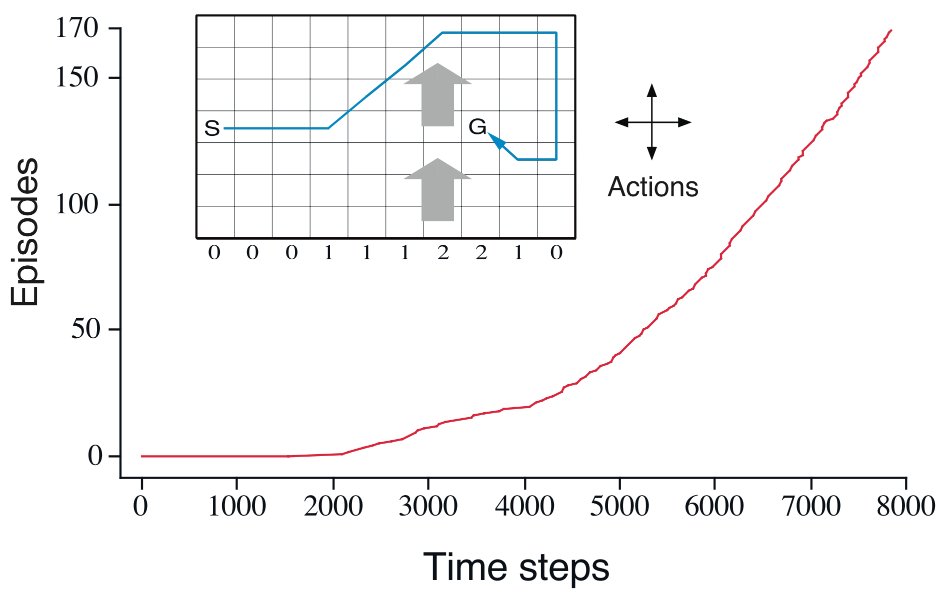
9 Sarsa Algorithm

Sarsa (on-policy TD control) for estimating $Q \approx q_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\mathit{terminal}, \cdot) = 0$

Loop for each episode:
  Initialize S
  Choose A from S using policy derived from Q (e.g., ε-greedy)
  Loop for each step of episode:
    Take action A, observe R, S'
    Choose A' from S' using policy derived from Q (e.g., ε-greedy)
    $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma Q(S', A') - Q(S, A)\big]$
    $S \leftarrow S'$; $A \leftarrow A'$;
  until S is terminal

10 Example: Windy Grid World



11 Q-Learning: Off-policy TD Control

Sarsa update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(S_t, A_t)\right]$$

Q-learning update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(S_t, A_t)\right]$$

Q-learning is off-policy because the value update is made using $\max_a$ rather than the a recommended by the policy being followed.

12 Q-Learning Algorithm

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\textit{terminal}, \cdot) = 0$

Loop for each episode:
    Initialize S
    Loop for each step of episode:
        Choose A from S using policy derived from Q (e.g., ε-greedy)
        Take action A, observe R, S'
        $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma \max_a Q(S', a) - Q(S, A)\big]$
        $S \leftarrow S'$
    until S is terminal